

Linear Equations: Guide

The big idea. A line is a **constant rate of change**. Two facts pin down any line: its **slope** m (how steep) and **one point** on it. Every equation form is just a different way to show those two facts.

WRITING THE EQUATION OF A LINE

- 1 Find the slope.** With two points, use $m = \frac{y_2 - y_1}{x_2 - x_1}$. Subtract in the *same order* on top and bottom.

2 Pick one point. Either given point works. Use the cleanest numbers.

3 Plug into point-slope. $y - y_1 = m(x - x_1)$. Keep the signs: a point $(-2, 5)$ gives $y - 5 = m(x + 2)$.
- 4 Simplify to $y = mx + b$.** Distribute m , then add y_1 to both sides.

5 Check with the other point. It must make the equation true. This catches sign slips.

6 Match the form asked for. Standard form is $Ax + By = C$; clear fractions so A, B, C are integers.

Most common mistakes: (1) Subtracting in different orders in the slope formula. (2) Using $m = \frac{x_2 - x_1}{y_2 - y_1}$ (run over rise). (3) Flipping the sign of the perpendicular slope but forgetting to flip the fraction, or the other way round. (4) Reading the slope from $3x + 4y = 12$ as 3 or -3 . Solve for y first.

WHICH FORM, WHICH JOB?

WHAT YOU ARE GIVEN	BEST STARTING FORM	WHAT TO DO
Slope and y -intercept	$y = mx + b$	Plug in m and b. Done.
Slope and any point	Point-slope	$y - y_1 = m(x - x_1)$, then simplify.
Two points	Slope, then point-slope	Find m first. Then use either point.
Standard form $Ax + By = C$	Solve for y	$m = -\frac{A}{B}$. y-intercept $= \frac{C}{B}$.
Parallel / perpendicular	Change the slope	Parallel: same m. Perpendicular: $-\frac{1}{m}$.

FORMULA BANK

SLOPE Formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ Meaning $m = \frac{\text{rise}}{\text{run}}$ Up to the right: $m > 0$. Down to the right: $m < 0$. Flat: $m = 0$. Vertical: undefined.	THREE FORMS Slope-intercept $y = mx + b$ Point-slope $y - y_1 = m(x - x_1)$ Standard $Ax + By = C$	SPECIAL LINES Horizontal: $y = k$. Slope 0. Vertical: $x = h$. Slope undefined (no y form). Two points with the same y make a horizontal line. The same x makes a vertical line.
PARALLEL & PERPENDICULAR Parallel $m_1 = m_2$ Perpendicular $m_1 m_2 = -1$ Perpendicular slope: flip the fraction and change the sign. $\frac{2}{3} \rightarrow -\frac{3}{2}$.	INTERCEPTS y-intercept: set $x = 0$. x-intercept: set $y = 0$. Plot both, draw the line through them. Good for standard form.	LINEAR MODELS $y = mx + b$ in a word problem: $m = \text{rate}$ (per hour, per item). $b = \text{starting amount}$ (fee, deposit).

Eight problems that cover the exam

Same plan every time: slope, point, equation, check.

Walkthrough 1: Slope from two points

SAME ORDER

$(-2, 5)$ and $(4, -1)$. Find the slope.

1 SET UP $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1 - 5}{4 - (-2)}$

2 CAREFUL Subtracting a negative: $4 - (-2) = 6$.

3 SIMPLIFY $m = \frac{-6}{6} = -1$

Walkthrough 2: Equation from two points

TWO STEPS

Write the line through $(1, 3)$ and $(4, 9)$.

1 SLOPE $m = \frac{9 - 3}{4 - 1} = \frac{6}{3} = 2$

2 SETUP Use $(1, 3)$: $y - 3 = 2(x - 1)$

3 SIMPLIFY $y - 3 = 2x - 2$, so $y = 2x + 1$

4 CHECK $(4, 9)$: $2(4) + 1 = 9$ ✓

Walkthrough 3: Standard form

SOLVE FOR Y

Find the slope and both intercepts of $3x + 4y = 12$.

1 SOLVE $4y = -3x + 12$, so $y = -\frac{3}{4}x + 3$

2 READ Slope $-\frac{3}{4}$. y -intercept 3.

3 X-INT. Set $y = 0$: $3x = 12$, so $x = 4$.

Walkthrough 7: A linear model

RATE AND STARTING AMOUNT

A plumber charges a \$45 trip fee plus \$30 per hour. Write the cost C for h hours, then find h when the bill is \$255.

1 MODEL Rate = 30, start = 45: $C = 30h + 45$

2 SET EQUAL $30h + 45 = 255$

3 SOLVE $30h = 210$, so $h = 7$ hours.

Walkthrough 8: Horizontal and vertical

SPECIAL LINES

Write the line through (a) $(3, -2)$ and $(3, 5)$, (b) $(-4, 2)$ and $(6, 2)$.

1 PART (A) Same x : a **vertical** line. $m = \frac{7}{0}$ is undefined.
Equation: $x = 3$

2 PART (B) Same y : **horizontal**. $m = \frac{0}{10} = 0$. Equation:
 $y = 2$

3 WATCH OUT **Zero** slope (flat) is not the same as **undefined** slope (vertical).

Walkthrough 4: Parallel & perpendicular

SLOPE

The line $4x - 2y = 6$ is given. Write the line through $(4, 1)$ that is (a) parallel and (b) perpendicular to it.

1 SLOPE $-2y = -4x + 6$, so $y = 2x - 3$ and $m = 2$.

2 PARALLEL Same slope: $y - 1 = 2(x - 4)$, so $y = 2x - 7$

3 PERP. SLOPE Flip and negate 2: $-\frac{1}{2}$.

4 PERP. LINE $y - 1 = -\frac{1}{2}(x - 4)$, so $y = -\frac{1}{2}x + 3$

Walkthrough 5: Graph an equation

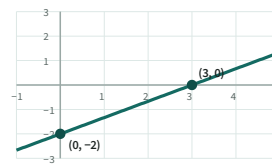
RISE OVER RUN

Graph $y = \frac{2}{3}x - 2$.

1 START The y -intercept is -2 . Plot $(0, -2)$.

2 MOVE Slope $\frac{2}{3}$: up 2, right 3. Plot $(3, 0)$.

3 DRAW Connect and extend both ways.



Two points are enough

Walkthrough 6: Equation from a graph

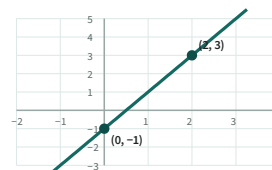
TWO CLEAN POINTS

Find the equation of the line shown.

1 POINTS It crosses the y -axis at $(0, -1)$ and passes through $(2, 3)$.

2 SLOPE $m = \frac{3 - (-1)}{2 - 0} = 2$

3 WRITE IT $b = -1$, so $y = 2x - 1$



Pick points on grid crossings